

EPREUVE DU 18 JANVIER 2021

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space. Let $T > 0$. Let (W_t) be a standard $\mathbb{R}$ -valued $(\mathcal{F}_t)$ -Brownian motion, where $(\mathcal{F}_t)$ is a filtration fulfilling the usual conditions. Let $(a_s), (b_s)$ be bounded progressively measurable processes.

1. We set $M := \int_0^\cdot a_s dW_s$. Show that there is a constant $c > 0$ such that

$$E(\exp(c \sup_{t \leq T} M_t^2)) < \infty.$$

Hint. If M is a continuous local martingale vanishing at zero, there is a standard Brownian motion $\mathcal{W}$ such that

$$M_t = \mathcal{W}_{[M, M]_t}, t \in [0, T],$$

(given without required justification).

2. Let $(M_t)_{t \in [0, T]}$ be a continuous $(\mathcal{F}_t)$ -local martingale. Let $\varepsilon, \delta > 0$. We introduce the random time

$$\tau := \begin{cases} \inf\{t \in [0, T] \mid [M, M]_t > \delta\} & \text{if } \{\} \neq \emptyset \\ \infty & : \text{otherwise.} \end{cases} \quad (1)$$

- (a) Explain why τ is a stopping time and $\{\tau \leq T\} \subset \{[M, M]_T \geq \delta\}$.
- (b) Show that

$$P\left(\max_{t \in [0, T]} |M_t| \geq \varepsilon\right) \leq \frac{E(\delta \wedge [M, M]_T)}{\varepsilon^2} + P([M, M]_T \geq \delta).$$

3. Let $\hat{b}, \hat{a} : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ two bounded Borel functions such that

$$\hat{b}(t, x) = E(b_t | X_t = x), \quad \hat{a}(t, x) = \sqrt{E(a_t^2 | X_t = x)}, \quad t \in [0, T], \quad x \in \mathbb{R}.$$

Let X be an stochastic process defined by

$$X_t = X_0 + \int_0^t a_s dW_s + \int_0^t b_s ds. \quad (2)$$

(a) We also suppose $\hat{a} \geq K > 0$ on $[0, T] \times \mathbb{R}$. Justify that the $SDE(\hat{a}, \hat{b})$, i.e.

$$\hat{X}_t = X_0 + \int_0^t \hat{a}(s, \hat{X}_s) dW_s + \int_0^t \hat{b}(s, \hat{X}_s) ds, \quad (3)$$

admits existence and uniqueness in law.

We denote by $\mathcal{P}(\mathbb{R})$ the space of Borel real probabilities. Let ν be a real probability. A function $\mu : [0, T] \rightarrow \mathcal{P}(\mathbb{R})$ is said to be a solution in the sense of (Schwartz) distributions of

$$\begin{aligned} \partial_t \mu(t) &= \partial_{xx}^2 \left(\frac{\hat{a}^2}{2}(t, \cdot) \mu(t) \right) - \partial_x (\hat{b}(t, \cdot) \mu(t)), \\ \mu(0) &= \nu, \end{aligned} \quad (4)$$

if for every $\varphi \in C^\infty$ with compact support and $t \in [0, T]$ we have

$$\begin{aligned} \int_{\mathbb{R}} \varphi(x) \mu(t, dx) &= \int_{\mathbb{R}} \varphi(x) \nu(dx) + \int_0^t ds \int_{\mathbb{R}} \varphi''(x) \frac{\hat{a}^2}{2}(s, x) \mu(s, dx) \\ &+ \int_0^t ds \int_{\mathbb{R}} \varphi'(x) \hat{b}(s, x) \mu(s, dx). \end{aligned}$$

We define $u : [0, T] \rightarrow \mathcal{P}(\mathbb{R})$ (resp. $\hat{u} : [0, T] \rightarrow \mathcal{P}(\mathbb{R})$) such that

$$\begin{aligned} u(t, \cdot) &:= \text{Law of } X_t, \quad t \in [0, T], \\ \hat{u}(t, \cdot) &:= \text{Law of } \hat{X}_t, \quad t \in [0, T], \end{aligned}$$

where X (resp. $\hat{X}$) is defined by (2) (resp. is a solution of the SDE (3)) and X_0 is distributed according to ν .

- (b) Expanding $\varphi(\hat{X}_t)$ show that $\hat{u}$ is solution of (4).
- (c) Expanding $\varphi(X_t)$ show that u is also a solution of (4).
- (d) Conclude that, whenever there is at most one solution of (4) then X_t and $\hat{X}$ have the same distribution.

4. Let $\delta \geq 0, x_0 \geq 0$. We consider the SDE

$$\begin{cases} dX_t &= 2\sqrt{|X_t|}dW_t + \delta dt \\ X_0 &= x_0. \end{cases} \quad (5)$$

- (a) Explain why previous SDE (5) admits pathwise uniqueness.
- (b) Explain why a solution X of (5) is necessarily non-negative.
- (c) During the exercises session we have proved that $E(X_t) = x_0 + \delta t$. Deduce that

$$E(\sup_{t \leq T} |X_t|^2) < \infty. \quad (6)$$

- (d) Let $\hat{f} : [0, T] \times \mathbb{R}_+ \times \mathbb{R}^2, (t, x, y, z) \mapsto \hat{f}(t, x, y, z)$ Borel and bounded and $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}$ with linear growth. Consider a solution $u \in C^{1,2}([0, T] \times \mathbb{R}_+) \cap C^0([0, T] \times \mathbb{R}_+)$ with linear growth, of

$$\begin{cases} \partial_t u &+ 2x\partial_{xx}^2 u + \delta\partial_x u + \hat{f}(t, x, u, 2\sqrt{x}\partial_x u) = 0, & (t, x) \in [0, T] \times \mathbb{R}_+ \\ u(T, x) &= \phi(x). \end{cases} \quad (7)$$

We set $Y_t := u(t, X_t), t \in [0, T]$. Show that $E(\sup_{t \in [0, T]} |Y_t|^2) < \infty$.

- (e) Determine a Borel function $v : [0, T] \times \mathbb{R}_+ \rightarrow \mathbb{R}$ (in term of u) so that $Z_t = v(t, X_t), t \in [0, T]$ and (Y, Z) is a solution of the BSDE

$$Y_t = \phi(X_T) + \int_t^T \hat{f}(r, X_r, Y_r, Z_r)dr - \int_t^T Z_r dW_r, \quad t \in [0, T]. \quad (8)$$

- (f) Show in particular that $E(\int_0^T Z_t^2 dt) < \infty$.

Barème indicatif.

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- 2. 5
- 3. 6
- 4. 6

Seul document autorisé. Deux pages (ou une feuille) de résumé pour la première partie et deux pages (ou une feuille) de résumé pour la seconde partie.

Advanced Stochastic Calculus (Second part) (FQ308-OMI302b). END